



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Driven Adaptive Risk Prediction and Feedback-Based Learning for Software Change Management	
S. No.	University ID	Name
1	2420030045	Susmitha
2	2420030477	N. Amrutha Lekha
3	2420030597	Samhitha
4	2420030752	Manvitha Reddy
Name of the Guide		Swapna Reddy

Abstract

In agile, DevOps-driven development environments, software evolves through frequent, incremental changes as teams respond to shifting requirements, sprint cycles, and continuous stakeholder feedback. While this iterative pace enables rapid delivery, it also raises the risk that a single change introduces undetected defects, security weaknesses, or unexpected service disruption, especially when manual code review and static checklists remain the primary safeguards before deployment. Existing quality assurance practices typically assess each change in isolation, providing limited visibility into how factors such as change size, affected modules, historical defect trends, and dependency relationships combine to influence overall deployment risk.

This project proposes an adaptive software engineering approach to software change management that embeds AI-based risk prediction directly into the development lifecycle. Whenever a developer commits a change, the system extracts change-related attributes, including modified files, lines changed, affected modules, dependency relationships, prior defect history, and security-scan findings, and uses a machine learning model to estimate the risk level of the change, highlight potentially affected components, and recommend relevant tests. This prediction supports agile decision-making, informing sprint planning, code review prioritization, and release readiness, while also complementing DevSecOps practices by surfacing security-relevant risk signals alongside functional ones.

Central to the project is its adaptive learning loop, inspired by MLOps principles: after each deployment, the actual outcome, such as a successful release, a newly discovered defect, or a rollback, is recorded and fed back into the model, allowing risk predictions to improve continuously as the team's Git history and defect patterns grow, much as agile teams themselves adapt through iterative retrospectives. The system will be implemented using a Git-integrated CI/CD pipeline, with containerized deployment and monitoring providing the operational environment in which this feedback loop runs.

By combining agile development discipline, DevOps automation, and adaptive machine learning, the project offers a practical demonstration of how software engineering processes can become data-driven and self-improving, helping development teams make more informed, risk-aware decisions without disrupting existing agile workflows.

Signature of the Guide

HOD